

Course Code: 23ADE038		Course Title: Natural Language Processing Systems	
Course Category: Major		Course Level: Higher	
L:T:P(Hours/Week) 2:0:2	Credits:3	Total Contact Hours:60	Max Marks:100

Course Objectives:

The course is designed to provide learners with the skills to perform word and sentence recognition, an essential foundation in Natural Language Processing.

Module I Language Modeling, Word Analysis, and Syntax Parsing 15 Hours

Language Modeling: Knowledge in Speech and Language Processing – Ambiguity – Regular Expressions Finite State Automata – Morphology –Finite State Transducers -Large Language Models (LLMs): Architecture, pre-training, and fine-tuning- Word and Sentence Tokenization – Detecting and Correcting Spelling Errors –Minimum Edit Distance.

Word Level Analysis: N-grams –Unsmoothed N-grams – Perplexity – Smoothing –Word Classes – Part-of-Speech Tagging – Rule-based, Stochastic and Transformation based tagging – Evaluation and issues in PoS tagging – Markov chains – Hidden Markov Model – Forward, Viterbi, Forward- Backward algorithms.

Syntax Analysis: Context-Free Grammars –Grammar rules – Treebanks - Dependency Grammars – Parsing as Search – Ambiguity –Dynamic Programming parsing – Partial parsing – Probabilistic CFG – Probabilistic CKY parsing –Probabilistic Lexicalized CFGs.

Module II Semantics and Pragmatics in Natural Language Processing 15 Hours

Semantics: First Order Logic –Description Logics – Syntax-driven semantic analysis – Word Senses – Relations between Senses –Semantic roles — Word Sense Disambiguation: Supervised, Dictionary & Thesaurus methods – Word Similarity: Thesaurus and Distributional methods - LLMs for Semantic Analysis: Using transformer models like BERT, GPT for semantic roles and disambiguation.

Pragmatics and Applications: Discourse segmentation – Text Coherence – Reference, Anaphora and Co-reference resolution –Named Entity Recognition – Relation Detection and Classification – Information Retrieval – Factoid Question Answering – Summarization.

List of Experiments:**30 Hours**

1. Download nltk and packages. Use it to print the tokens in a document and the sentences from it. Include custom stop words and remove them and all stop words from a given document using nltk or spaCY package
 2. Implement a stemmer and a lemmatizer program.
 3. Implement a simple Part-of-Speech Tagger
 4. Write a program to calculate TFIDF of documents and find the cosine similarity between any two documents.
 5. Use nltk to implement a dependency parser
 6. Implement a semantic language processor that uses WordNet for semantic tagging.
- The given experiments to be implemented using Python with NLTK /spaCy /scikit learn.

Course Outcomes	Cognitive Level
At the end of this course, students will be able to:	
CO1: Analyze the core principles of language modeling, word tokenization, morphology, and finite-state methods to address ambiguity in natural language.	Analyze
CO2: Apply probabilistic techniques like N-grams, HMMs, and POS tagging algorithms to perform effective word-level analysis and tagging.	Apply
CO3: Evaluate syntactic structures and parsing techniques using CFGs, dependency grammars, and probabilistic models for accurate syntactic analysis.	Evaluate
CO4: Create semantic and pragmatic applications using LLMs and techniques such as semantic role labeling, word sense disambiguation, NER, and summarization.	Create

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	-	2	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	-	1	-	-	-	-	-	-	2	
CO3	-	-	-	3	-	-	-	-	-	-	-	-	-	-
CO4	-	-	3	-	-	-	-	-	-	-	-	2	-	2

High-3; Medium-2;Low-1

Text Book(s):

- T1. Daniel Jurafsky and James H. Martin, "Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition with language models", 3rd Edition, Pearson, 2020 (Module I)
- T2. U. S. Tiwary and Tanveer Siddiqui, "Natural Language Processing and information Retrieval", Oxford University Press, 2008. (Module I, II)

Reference Book(s):

- R1. Nitin Indurkha, Fred J. Damerau, "Handbook of Natural Language Processing", Second edition, Chapman & Hall/CRC: Machine Learning & Pattern Recognition, Hardcover, 2010
- R2. Mohamed Zakaria Kurdi "Natural Language Processing and Computational Linguistics: Speech, Morphology and Syntax (Cognitive Science)", ISTE Ltd., 2016

Web References:

- 1. <https://nptel.ac.in/courses/106101007/>
- 2. <https://nlp.stanford.edu/software/>
- 3. <https://www.ibm.com/topics/natural-language-processing>